

GE261004 DESIGN THINKING

Unit IV – Prototyping and Testing · Presentation 3 of 6

Low-Fidelity Prototyping

What rough buys you · paper, cardboard, clay and role play · Wizard of Oz · prototyping behaviour · building something usable in an hour

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In this presentation

- What low fidelity buys you
- Six ways to build rough
- Four materials and their uses
- Prototyping behaviour
- Wizard of Oz
- Role play and bodystorming
- Building it in an hour
- Rules for a lo-fi build



1

Building rough

Cheap enough to change, ugly enough to criticise

What low fidelity buys you

- Speed: an hour instead of a week, so you can test three ideas, not one
- Honesty: people criticise cardboard freely, and will not criticise something that looks finished
- Focus: a rough model directs attention to the idea, not the surface
- Detachment: you can throw away something you made in an hour

KEY POINT Polish buys politeness. If you want the truth, hand somebody something that is obviously unfinished.

Six ways to build rough

1

Paper

fast layouts and
screen flows

2

Cardboard

shape, size and
where the hands go

3

Clay or foam

form you can hold
and reshape

4

Role play

services and
conversations

5

Wizard of Oz

behaviour before
the software exists

6

Click-through

screens linked by
hand-drawn buttons

Pick by what you need to learn, not by what is on the table.

Four materials and what they are good for

Paper and card

Screens, forms, layouts and anything flat. Cut and redraw between users.

Cardboard and tape

Size, shape and where the hands go. The only way to feel whether it is too heavy.

Clay or foam

Form you hold: grips, handles, casings. Reshape it while the user is still talking.

Found objects

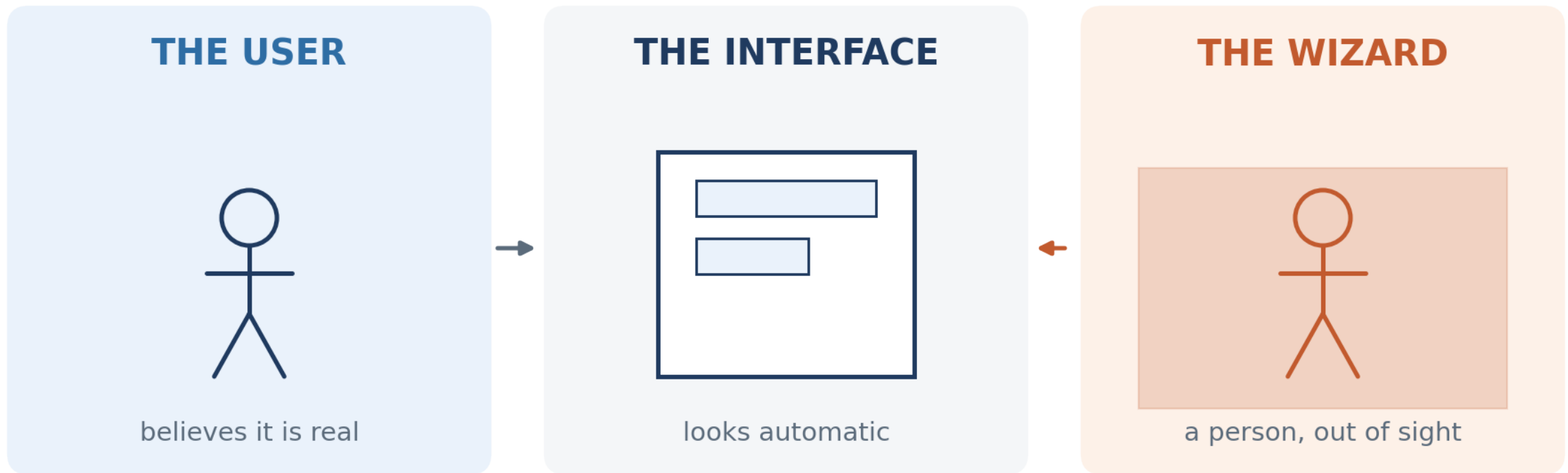
A bottle, a box, a phone standing in for the thing. Fastest of all, and usually enough.

2

Prototyping behaviour, not objects

When the thing you are testing is what happens, not what exists

The Wizard of Oz set-up



test the behaviour now; build the machinery only if the behaviour works

John Kelley named the method around 1980, while testing a natural-language system.

Wizard of Oz

- The user believes the system is automatic; a hidden person is producing the responses
- Kelley used it to test a natural-language calendar before the software existed
- It lets you test whether a behaviour is wanted, before you spend months building it
- Tell participants afterwards what happened – always, without exception

KEY POINT Build the machinery only after you know somebody wants what the machinery would do.

Role play and bodystorming

- **Role play** – act out the service: one plays the user, one the staff, one narrates
- **Bodystorming** – go to the real place and act it out there, with real furniture and real distance
- Both surface things no drawing shows: who stands where, who speaks first, what is awkward
- Props can be anything – a chair is a counter, a book is a machine

KEY POINT If your solution is a service, a drawing of it tells you almost nothing. Act it out instead.

Building it in an hour

- Decide the one question first and write it on the table
- Ten minutes deciding, thirty building, twenty testing on a real person
- Use whatever is within reach; a trip to buy materials has already cost you the hour
- Stop building the moment it can answer the question, even if it looks unfinished

ACTIVITY Sixty minutes, materials on the table, no shopping. Every team must test on one person from outside the team before the hour ends.

Rules for a lo-fi build

Do

- Write the question on it
- Build at real size
- Let the user break it
- Change it between users

Do not

- Add features nobody asked about
- Make it look finished
- Explain it before they try
- Defend it when it fails

KEY POINT The moment you start explaining your prototype, the test has ended and a sales pitch has begun.

In class: build the roughest version

- Take the riskiest question your team wrote in the first hour of this unit
- Build the roughest thing that could answer it, from what is on the table
- Test it on somebody from another team, saying nothing while they use it
- Write down the three things that surprised you

ACTIVITY One hour. Bring the prototype and the three surprises to the next session.

THINK ABOUT IT

What did your user do with your prototype that you did not expect?

- Which part did they try to use in the wrong way?
- What did they look for that was not there?
- Where did they hesitate, and what were they deciding?
- What would you change before the next person touches it?

Summary

- Rough buys speed, honesty, focus and the willingness to throw it away
- Paper, cardboard, clay and found objects each answer different questions
- Build at real size – scale and weight cannot be imagined
- Wizard of Oz tests a behaviour before the system exists
- Role play and bodystorming prototype services and conversations
- Write the question on the prototype and stop building when it can answer it

Next: Rapid prototyping

- From sketch to solid part
- What CAD adds, and what it costs you
- FDM, SLA and SLS compared
- Laser cutting, CNC and digital mock-ups